

Statement of Student Success

Educational Philosophy. I believe rigorous education is most effective when high expectations are paired with strong support and an environment in which students feel respected, are comfortable asking questions and expressing ideas, and have meaningful opportunities to succeed. I believe an effective academic community is built on individual dignity, mutual respect, open expression, and shared responsibility for creating an environment that supports learning and growth. Throughout my academic career, I have worked with students with different educational experiences, levels of preparation, academic goals, and approaches to learning. These experiences have reinforced my commitment to helping students develop purpose, belonging, and confidence, technical competence, intellectual independence, and the ability to engage constructively with others and contribute meaningfully to a collaborative academic community within and beyond the classroom.

Supporting Student Learning and Success. My teaching experience has shown me that students enter technically demanding courses with different levels of mathematical preparation, programming experience, confidence, and familiarity with technical material. I therefore combine high academic standards with clear communication, accessible explanations, regular feedback, and multiple opportunities for active engagement through office hours, discussions, and mentoring. At the University of Illinois Urbana-Champaign, I support students through classroom instruction, detailed course materials, and individual discussions that provide additional opportunities for guidance. I structure challenging material so that students understand not only how to apply a method, but why it works and how it connects to broader ideas and real-world problems. My goal is to help students develop the strong foundation and confidence needed to solve challenging problems independently.

Beyond formal coursework, I advise students about research opportunities, graduate school preparation, career planning, and professional development, particularly those exploring research or graduate education for the first time. I have also participated in the Society of Hispanic Professional Engineers (SHPE) National Convention and the Emerging Researchers National (ERN) Conference in STEM, supporting efforts to connect students with educational, research, and professional opportunities. These experiences have reinforced the value of making academic and professional pathways accessible.

Creating a Respectful Learning Environment. Effective learning depends not only on strong instruction but also on an environment in which students feel comfortable participating, asking questions, and engaging constructively with one another. An academic community should allow students to express ideas, disagree, and learn from different perspectives while maintaining respect for others. Faculty have an important responsibility for establishing these expectations.

As a Visiting Lecturer at the University of Maryland, Baltimore County, one experience that influenced my approach occurred when a student made a negative reference to a particular national identity during a class discussion. I addressed the issue immediately and emphasized the importance of maintaining a respectful learning environment. The experience reinforced my belief that intellectual openness and respectful discourse are complementary: students should feel free to contribute their ideas and perspectives while recognizing their responsibility to engage with others constructively.

As a Graduate Teaching Assistant at Purdue University, I similarly worked to promote effective collaboration in team-based assignments. We formed teams that brought together students with different experiences and approaches, allowing them to learn from one another while developing collaborative problem-solving skills. We also used student feedback to understand how teams were functioning. In one instance, students raised concerns about interactions within a project team. We responded promptly, worked with the students involved, reinforced expectations for respectful collaboration, and helped the team establish a productive working environment. The team subsequently completed its assignments successfully. These experiences strengthened my appreciation for listening to students, responding constructively to concerns, and establishing clear expectations for collaboration.

Mentoring and Advising. Mentoring has become one of the most meaningful components of my academic work. At Illinois, I mentor undergraduate, master's, and doctoral students through research advising, independent study, and structured mentoring activities. I lead a research group in which students work on projects involving algorithms, artificial intelligence, networks, and data-driven decision-making. These experiences allow students to develop technical skills, gain research experience, and explore future educational and professional pathways.

Students enter research with different levels of experience, and I adapt my mentoring accordingly. Early in a project, I provide structure and help students develop the necessary technical and scholarly foundations. As they progress, I encourage greater ownership of research questions, methodology, experimentation, and communication. I also advise students on graduate school preparation, career planning, and professional development. My goal is to help them develop the judgment and independence needed to pursue their longer-term academic and professional goals.

At Purdue, I served as a mentor in the GradTrack program, helping students interested in graduate education understand research opportunities, applications, and academic careers. As an Instructor with Johns Hopkins University's Center for Talented Youth, I worked with academically gifted students generally ages 12 to 15, strengthening my ability to adapt instruction and encourage intellectual curiosity. My teaching and mentoring have been recognized through the Graduate Mentor Award and Trailblazers in Engineering Fellowship at Purdue and the Stanley H. Pierce Award and Engineering Council Outstanding Advisor Award at Illinois.

Commitment to Student Success. I bring these principles into my teaching, mentoring, research group, and service. I maintain rigorous academic expectations while providing learning resources that support students with different levels of preparation. I encourage questions, intellectual curiosity, and respectful disagreement while establishing expectations for constructive interaction. I also establish clear expectations for responsible use of generative AI so that these tools support, rather than replace, students' analytical and problem-solving development.

As a research mentor, I create opportunities for undergraduate and graduate students to engage in research, adapting my guidance as they develop greater independence. I help students identify educational and professional pathways and develop the technical competence, intellectual confidence, communication skills, and professional judgment needed to pursue them successfully. I view student success not simply as performance in an individual course, but as the longer-term development of students as independent thinkers, collaborators, researchers, and professionals.